

STEMY — How The Website Works

Hello! This document explains my website project. Please read it slowly. If anything is not clear, please write your questions and send them to me. I will answer in writing. Writing is easier for both of us than video calls.

1. WHAT IS STEMY?

Stemy is a website for music artists. Artists upload their songs. The website makes the songs sound professional. This is called "mastering."

Like these companies:

- LANDR.com
- eMastered.com
- BandLab Mastering

Stemy will be similar but with two important differences:

1. **Better stem mastering** (explained later)
 2. **Built by a real producer** (me — Choo, with 30+ years of experience)
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2. THE TWO PLANS (PRICING)

Customers can pay one of two prices per month:

Basic Plan — \$9.99 per month

- Customer uploads ONE finished song file
- Customer picks a music style (Pop, Hip-Hop, R&B, Rock, Electronic, Acoustic)
- Website processes the song
- Customer downloads the finished master
- **Unlimited songs per month**

Pro Plan — \$24.99 per month

- Customer uploads SIX separate audio files (called "stems"):
 1. Drums
 2. Bass
 3. Vocals
 4. Keys (piano, synth)
 5. Guitar
 6. FX (sound effects)
- Customer can adjust each stem (volume, EQ, effects, auto-tune for vocals)
- Customer can upload a "reference song" for the system to match
- Website mixes all stems together AND masters
- Customer downloads finished song
- **Unlimited songs per month**

Free Trial

- Both plans give the customer **7 free days** to try
 - After 7 days, the credit card is charged automatically
 - Customer can cancel anytime
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3. WHAT THE WEBSITE NEEDS TO DO

A) Customer makes an account

- Customer types: name, email, password
- OR customer logs in with Google
- Email is sent to verify the email address
- Customer can reset password if forgotten

B) Customer pays subscription

- Customer picks Basic (\$9.99) or Pro (\$24.99)
- Credit card is collected through Stripe (a payment company)
- 7-day free trial — no charge during trial
- After 7 days, charge automatically every month
- Customer can cancel anytime in their account page

C) Customer uploads a song (Basic Plan)

- Customer selects an audio file from their computer
- File types accepted: WAV, MP3, FLAC, M4A, AIFF
- Maximum file size: 100 MB
- Customer chooses a music style
- Customer clicks "Master This Track" button
- Website processes the song (takes about 90 seconds)
- Customer can download the finished file as WAV (24-bit) and MP3

D) Customer uploads stems (Pro Plan)

- Customer uploads up to 6 separate files (one for each stem)
- Each stem appears as a "channel" on a virtual mixing console
- Customer can adjust each channel:
 - Volume (fader)
 - EQ (4 knobs: HI, HMID, LMID, LOW)
 - Compression (3 knobs: THR, RAT, ATK)
 - Effects (REV, DLY, SAT)
 - For vocals only: AUTO-TUNE with key selection
 - Mute / Solo buttons
- Optional: Customer uploads a "reference song"
- Website analyzes the reference and matches the customer's mix to it
- Customer clicks "Master Track" button

- Website mixes everything and applies mastering
- Customer downloads finished file

E) Customer can add information to the file

- Cover artwork (image)
- Song title
- Artist name
- Album name
- Year
- Genre
- Track number
- ISRC code (a music industry ID number)
- Composer name
- Copyright information

This information is “embedded” in the audio file (called metadata).

F) Customer’s account page

- See current subscription
- See past masters (download history)
- Cancel or change plan
- Update name, email, password
- Upload profile picture

G) Email notifications

The website sends emails for:

- Welcome (after signup)
- Email verification
- Password reset
- “Your master is ready”

- Receipt after payment
 - Trial ending in 3 days
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4. WHAT THE FRONTEND ALREADY HAS (THE PARTS YOU CAN SEE)

I already built the design and visual parts of the website. The HTML file has:

✓ Home page with hero, features, comparison, pricing, FAQ ✓ Sign up page ✓ Log in page ✓ Forgot password page ✓ My Account page (with avatar upload, settings, subscription info, download history) ✓ Studio page with two tabs (Quick Master and Pro Mixing Console) ✓ About Choo page (about the founder) ✓ Pricing modal (where customer enters credit card) ✓ All animations and visual effects

Your job is to make these things actually work by connecting them to a real backend.

5. WHAT THE BACKEND NEEDS TO DO (YOUR JOB)

Build these systems:

1. **User database** Store: name, email, password (hashed), avatar, signup date, subscription status
2. **Authentication system** Login, logout, password reset, email verification, Google login
3. **Stripe payment integration**
 - Subscribe to plan
 - Charge monthly
 - Cancel subscription
 - Update card
 - Handle failed payments
4. **File upload system** Customer uploads audio files. Files saved to cloud storage (Cloudflare R2 or AWS S3).
5. **Audio mastering engine** This is the most important part. Two options:

Option 1: Use a third-party API (faster to build)

- Auphonic API
- Or similar service

Option 2: Build it yourself (more work)

- Use Python with the “pedalboard” library (made by Spotify) or
- Use sox (audio processing tool)
- Apply: high-pass filter, EQ (4 bands), saturation, compression, stereo widener, brickwall limiter
- Output target: -14 LUFS, -1 dBTP (this is the standard for Spotify, Apple Music, etc.)

6. **Email system** Use Resend or SendGrid to send transactional emails

7. **Job queue** Mastering takes 90 seconds. Don’t make the customer wait — process in the background. Use BullMQ (Node.js) or Celery (Python) with Redis.

6. EASIEST TECH STACK TO USE

If you want to build this fast:

- **Backend language:** Node.js (Express) OR Python (FastAPI) — your choice
- **Database:** PostgreSQL (use Supabase or Neon for managed Postgres)
- **Authentication:** Supabase Auth (free tier, includes email + Google login)
- **Payments:** Stripe
- **File storage:** Cloudflare R2 (cheaper than AWS)
- **Email:** Resend
- **Hosting backend:** Railway or Render
- **Hosting frontend:** Vercel
- **Mastering:** Start with Auphonic API, switch to custom DSP later if needed

7. IMPORTANT NOTES

1. **The frontend HTML file is one single file** — about 614 KB. You can split it later if needed, but for now keep it as one file.

2. **The frontend uses `localStorage` for testing** — this is fake data for demo only. You will replace this with real API calls to your backend.

Look for these in the code:

- `localStorage.setItem('esLoggedIn', '1')` → replace with real login
- `localStorage.getItem('esUserName')` → replace with real user data from API
- `localStorage.getItem('esUserAvatar')` → replace with avatar URL from your storage

3. **Genre preset values for mastering** are in the HTML file. Search for `QMX_GENRES` in the code — these are the EQ, compression, and saturation settings for each music style. Use these values in your backend mastering engine so the result matches what the customer hears in the preview.

4. **Before launching, I need legal pages:**

- Privacy Policy
- Terms of Service
- Refund Policy

Stripe will not activate the account without these pages.

8. HOW WE WILL COMMUNICATE

- I prefer **written messages** instead of video calls. Please write your questions in messages so I can read them carefully.
- If you write in your language and translate to English using Google Translate, that is okay. I will do the same if needed.
- Please tell me if anything is not clear in this document.
- Please tell me how many days/weeks each part will take.
- Please send updates every 2-3 days so I know what is finished.

9. QUESTIONS I WANT YOU TO ANSWER

Please answer these questions in writing before we start:

1. Do you understand all parts of this document?
 2. Which parts do you have questions about?
 3. Have you built a subscription website with Stripe before? Show me an example.
 4. Have you worked with audio processing before? If yes, which tools?
 5. Do you prefer Node.js or Python?
 6. How many weeks will the full backend take?
 7. What is your total price for this project?
 8. How many hours per week will you work on this?
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10. PROJECT NAME AND CONTACT

Project: Stemy **Website (when ready):** stemy.com (or similar) **My name:** Choo **My role:** Founder **Contact:** [your email and WhatsApp here]

Please send all questions to me in writing. I will respond within 24 hours.

Thank you!